



Effects of Asymmetries in Merging Dual Jet Streams on Downstream Velocity Profiles and Turbulent Flow Characteristics

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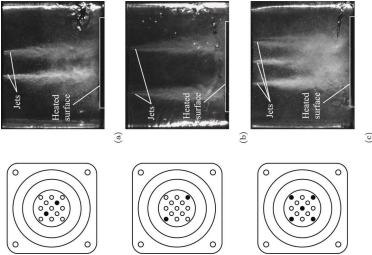


Figure 2: Parallel jets in three different configurations impinging upon a heated surface [1].

INTRODUCTION

- Applications of Dual Parallel Jets



Figure 1: Twin jet configuration for NASA aircraft [2].



INTRODUCTION

Symmetrical

- Merge point (mp)
- Combined point (cp)
- Turbulence behavior

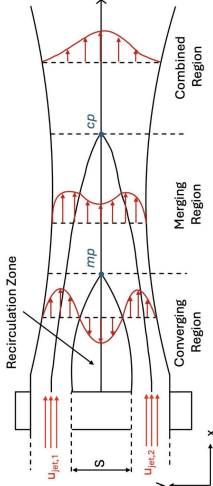


Figure 3: Dual-jet stream characteristics from parallel jets. Velocities from each jet are equal, $u_{jet,1} = u_{jet,2}$. Red arrows depict flow velocity profiles [3].



INTRODUCTION

Expectations

- Merge point (mp)
 - $\sim 6D$
- Combined point (cp)
 - $\sim 16D$

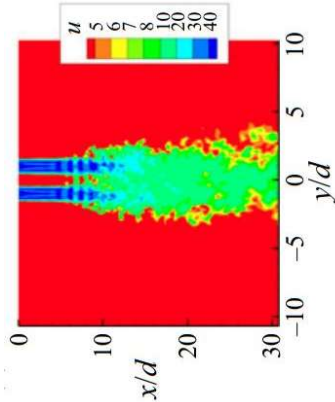


Figure 4: Velocity visualization for a Reynolds number of 25000 and an S/D value of 2 [4].



INTRODUCTION

Asymmetrical

- Merge point (mp)
- Combined point (cp)
- Turbulence behavior

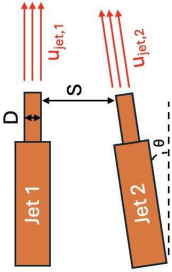


Figure 4: Asymmetric dual jets with jet 2 incident on jet 1 an angle of θ .



INTRODUCTION

Objective

Determine how introducing an asymmetric angle on parallel dual jets of the same velocity and Reynolds Number affects the locations of the merge point, combined point, and turbulence characteristics downstream.

Expectations

- Closer mp and cp locations upstream to the jet exits, with sooner turbulent flow development for asymmetric jets as opposed to symmetrically parallel jets of the same Reynolds Number.
- Angled velocity profile development for combined flow.



EXPERIMENTAL SETUP

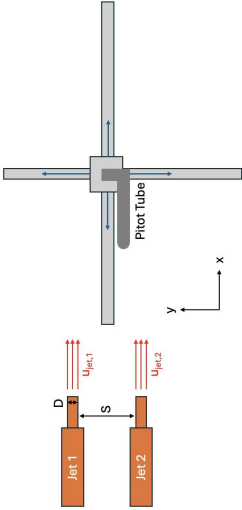


Figure 5: Two parallel round free jets with diameter D are set a distance S apart. The velocities of the jets, $u_{jet,1}$ and $u_{jet,2}$ are measured by a pitot tube that is able to move across both the x and y directions via a gantry carriage system.



EXPERIMENTAL SETUP

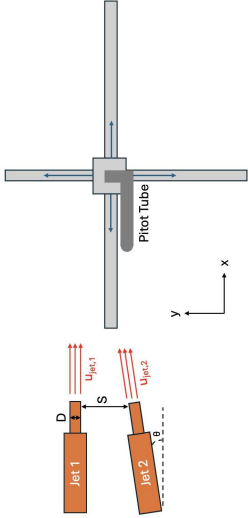
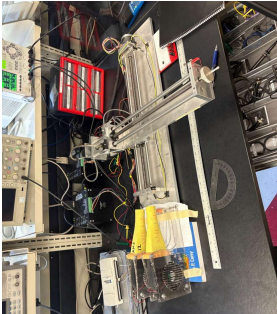
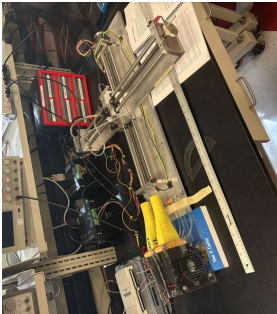


Figure 6: Experimental setup with one round free jet tilted at an angle θ . Incident angles are measured at positive θ (converging).



$$p_t - p_s = 1/2 \rho U^2$$
$$U = \sqrt{\frac{2(p_t - p_s)}{\rho}}$$
$$R_e = \frac{\rho U D}{\mu}$$

RESULTS

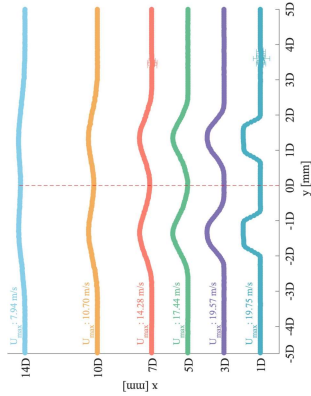


Figure 7: Velocity profile of parallel jets with centerline illustrating jet symmetry.

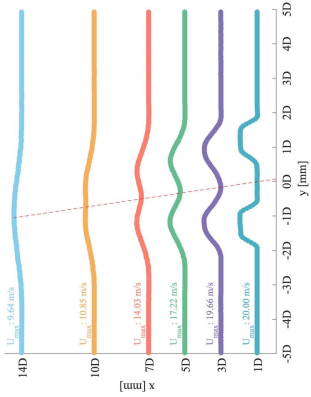


Figure 8: Velocity profile of dual jets with one angled. Centerline illustrates asymmetry with an angle of 4.59°.

RESULTS

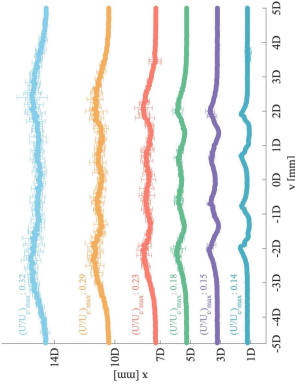


Figure 9: Turbulence intensities for parallel jets

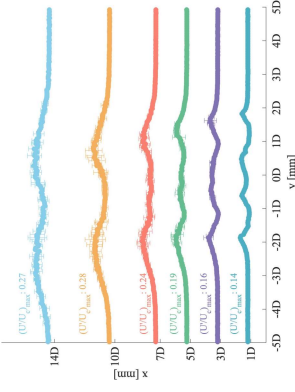


Figure 10: Turbulence intensities for dual jets with one angled at 10°.



RESULTS

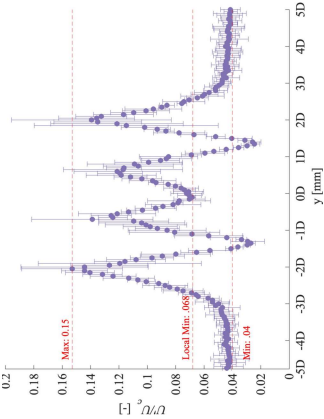


Figure 11: Representative trace for turbulence intensity of parallel dual jets at 3D.



CONCLUSION

Jet Configuration	Merge Point	Combined Point
Parallel	~5D	>14D
Angled	~1-3D	~7-10D



REFERENCES

[1] M. V. P. Carneiro and J. R. Barbosa Jr., "A comparison of parallel and colliding jet arrays in a compact vapour compression heat sink for electronics cooling," *Applied Thermal Engineering*, vol. 195, p. 117217, Aug. 2021. [Online]. Available: <https://doi.org/10.1016/j.applthermaleng.2021.117217>

[2] B. Henderson and R. Bozak, *Twin Jet*, presented at the Acoustics Technical Working Group, Oct. 21–22, 2010. NASA, May 2010. [Online]. Available: <https://ntrs.nasa.gov/api/citations/20100042248/downloads/20100042248.pdf>

[3] Murphy, Paula J., Sajjad Alimohammadi, and Séamus M. O'Shaughnessy. 2024. "Two-Dimensional Dual Jets—A Comprehensive Review of Experimental and Numerical Analyses" *Energies* 17, no. 17: 4487. <https://doi.org/10.3390/en17174487>

[4] T. M. Taddesse and J. Mathew, "Development and scaling of turbulent, twin round jets," *J. Fluid Mech.*, vol. 939, A13, 2022, doi: 10.1017/jfm.2022.193.



QUESTIONS?